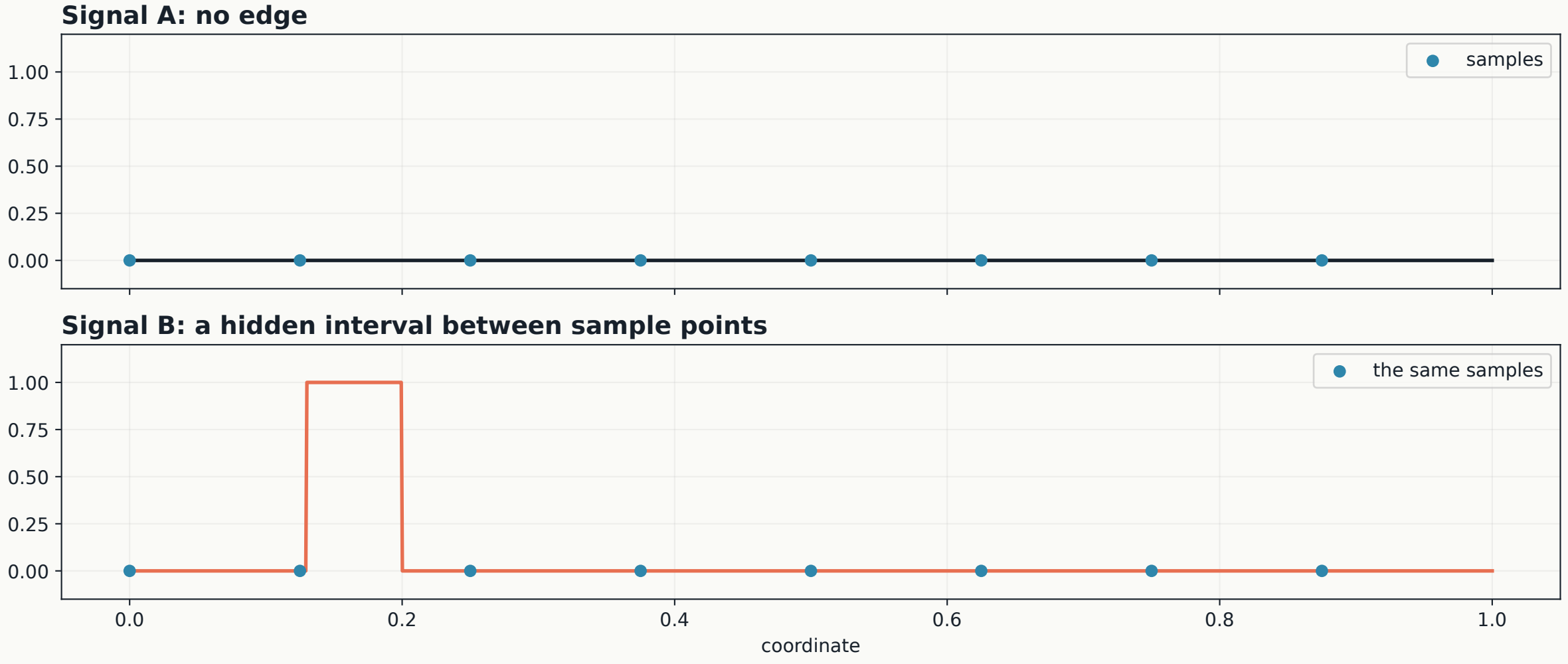No method can recover arbitrary edges from a noninjective measurement map